Gabriel Rodriguez

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EDUCATION

California State University Fullerton | Fullerton, CA **Expected December 2026**

Bachelor of Science in Computer Science, Minor in Data Science

Citrus College | Glendora, CA **June 2024**

Associate of Science in Computer Science

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Science & Big Data, Computational Bioinformatics

TECHNICAL SKILLS

Programming Languages: Python, SQL

Libraries/Frameworks: PyTorch, Torchvision, OpenCV, MONAI, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Concepts: Machine learning, Deep Learning, Multimodal Learning, Computer Vision, Object Oriented Programming, Feature Engineering, Prompt Engineering, Agile/Scrum

Tools: AWS (S3, SageMaker), CUDA, Git/GitHub, PyCharm, Visual Studio, Jupyter Notebook, Google Colab

PROFESSIONAL PROJECTS

MRI Brain Tumor Classifier (CNN) **November 2025**

- Designed and trained a **deep learning model** (**CNN**) for **medical image classification**, classifying brain tumors (glioma, meningioma, pituitary), achieving **89% accuracy** and **90% recall** on tumor-positive cases
- Enhanced classification **precision** by **11%,** leveraging a **Convolutional Block Attention Module** (**CBAM**) for **spatial/channel** focus and validated model interpretability with **Grad-CAM** visualizations aligned to tumor regions
- Minimized overfitting using **regularization** techniques incorporating **dropout**, **weight decay**, and a **learning rate scheduler** reducing validation loss by **20% on cross-validation test set**

Diabetes Prediction Model (ANN) **January 2025**

- Engineered an **Artificial Neural Network** in **PyTorch** to predict diabetes via patient **electronic medical records** (**EMR**), achieving **80% overall accuracy** and **92% recall** on diabetic cases
- Optimized patient data preparation, applying **Pandas** and **normalization** techniques, enhancing model convergence and boosting **recall** by **8%**
- Tuned class imbalance handling with **stratified splitting** and **class-weighted loss**, increasing **F1-score** for diabetic predictions by **10%**, ensuring high sensitivity for early diagnosis

RESEARCH EXPERIENCE

CSU Student Research Competition | California State University, San José **April 2026**

- **Represented California State University, Fullerton** at the **CSU Student Research Competition** developing a **semantic segmentation pipeline** for **Spontaneous Retinal Venous Pulsation** (**SVP**) detection
- Processed an **ophthalmic video dataset** (**635 videos, 130K+ frames**) using **data augmentation, temporal modeling**, and **GPU-accelerated training** to improve robustness under limited labeled data
- Extracted quantitative SVP biomarkers (vessel diameter and area change over time) to support **early, accessible detection** of **glaucoma** and **elevated intracranial pressure** (**ICP**)

Data Science and Machine Learning Club | California State University, Fullerton **January 2026**

- **Co-authored a research paper** and **led a team of 3 students** to develop a **multimodal deep learning pipeline** for **breast cancer T-stage classification** integrating **clinical data, radiomic features**, and **3D breast MRI volumes**
- Implemented a **scalable MONAI-based preprocessing** and training pipeline (**ROI extraction, voxel resampling, normalization**) achieving **97% F1-score on T-stage 4 patients**
- Architected a **late fusion 2.5D CNN with a ResNet backbone**, using stacked MRI slices to capture volumetric context and improving **T-stage 3 recall to 87%**